

Standard Operating Procedure (SOP) v4.0: DIW 3D Printing Calibration via a Per-Shape Test Suite

One STL per benchmark geometry, swept over $(v_{\text{print}}, \text{EM})$ at fixed slicer settings, paired with a flash+no-flash iPhone photograph for downstream image-based scoring.

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What changed from v3.0.

v3.0 collapsed the C1–C4 calibration battery into a single **Consolidated Calibration Coupon (CCC_v1)**. v4.0 returns to a *per-shape benchmark suite*: each printability quantity is measured on its own purpose-built specimen, swept over a $(v_{\text{print}}, \text{EM})$ grid. The shift is motivated by:

- **Reviewer-familiar geometries.** Hollow cubes, towers, 0/90 scaffolds, raster (S-test) and bridges are the standard benchmarks in the DIW literature. Reporting on these directly is defensible without explaining a one-off coupon design.
- **Quantitative image analysis at the shape scale.** Each shape isolates one failure mode, which simplifies the image-processing pipeline (Section 6) and gives a single *printability value* per shape per operating point.
- **Direct slicer authoring.** G-codes are sliced once per shape in PrusaSlicer and reused across inks. EM and v_{print} are swept by re-slicing, not by hand-baking E values.

Photography protocol (new in v4). Two iPhone photographs per G-code, taken from the same mounted position — one with the built-in flash, one without. Substrate background is either **black** or **blue** matte material (subfolded).

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1 Why a per-shape suite

KEY IDEA

A single geometry cannot simultaneously isolate line width, infill quality, layer-on-layer stacking, vertical stability, overhang behaviour, and pore fidelity. Each of these failure modes *is* a shape. The community-standard benchmark geometries (hollow cube, raster/S, scaffold 0/90, tower, bridge) were designed to do exactly that.

CCC_v1 (v3.0) prioritised throughput — one specimen per operating point — at the cost of measuring four heterogeneous quantities on the same print. v4.0 reverses that trade. Each benchmark is sliced once, swept over (v_{print} , EM), photographed under controlled lighting, and scored by a per-shape metric. The cost is roughly $5\times$ the number of prints, offset by:

- Direct comparison against published bioprinting benchmarks.
- One photograph → one quantitative score per shape, through the imaging pipeline of Section 6.
- No bespoke CAD — all geometries are public-domain benchmarks already used in the field.

2 Shape inventory

2.1 Folder organisation under Impressao 3D/

```

Impressao 3D/
  Area-Cube/ <- STL + 15 gcodes (Vp x Fr sweep)
  Bridge/ <- STL only
  Cubes/ <- STL only
  Misc/Pot/ <- STL + gcodes (in progress)
  Misc/Star/ <- STL + gcodes (in progress)
  S-Test/ <- STL + 20 gcodes (Vp x Flow sweep)
  SF5.5/
    Black/ <- iPhone photos on BLACK substrate
    Blue/ <- iPhone photos on BLUE substrate
  Scaffold 0-90/ <- STL + gcodes (in progress)

```

Shape	STL location	Quantifies	Failure modes
Hollow cube	Cubes/cubes.stl	Wall fidelity, corner sharpness, perimeter continuity	Wall collapse, corner rounding
Raster / S-test	S-Test/Teste de Printabilidade Raster.stl	Printability index (line straightness and uniformity in a raster pattern)	Under/over-extrusion, stringing
Area cubes (full-fill, varying height)	Area-Cube/Modelo Area Cubos.stl	Infill density, layer adhesion, top-surface flatness	Inter-line gaps, top-surface roughness
Tower 20 mm	Tower-20mm/	Short-tower stability, base bulge	Buckling onset, base spreading
Tower 40 mm	Tower-40mm/	Tall-tower buckling limit	Lateral collapse, base creep
Scaffold 0/90	Scaffold 0-90/	Pore fidelity, layer-on-layer registration	Pore closure, strand sag into pore
Bridge / suporte	Bridge/Modelo Suporte.stl	Bridging / overhang	Mid-span sag, dropout
Misc (Pot, Star, Pyramid, Maze)	Misc/	Complex-geometry validation	Geometry-specific (e.g. thin-feature failure)

Table 1: Benchmark shape inventory. STLs sliced once in PrusaSlicer, swept across (v_{print} , EM). “Failure modes” indicate the dominant artefact each shape is designed to expose.

```
Tower-20mm/ <- STL + gcodes (in progress)
Tower-40mm/ <- STL + gcodes (in progress)
OLD/ <- CCC_v1 archival (do not use)
```

The SF5.5/ subfolder is an ink identifier (here: C15 + sunflower-oil nanoemulsion @ 5.5%). Black/ and Blue/ are *substrate background colours*, not ink colours. The same operating point is photographed against both backgrounds when contrast against one background is degenerate (e.g. white ink fades against a high-key flash on blue but reads on black). Subsequent inks introduce a sibling subfolder (e.g. C15/, Bozano/) at the same level as SF5.5/.

3 Naming convention for G-codes

Token	Meaning	Sweep values currently in folder
VpN	v_{print} (print-head speed) in mm/s. Not piston velocity.	$N \in \{5, 7, 10, 15, 30\}$
FrX or FlowX	Extrusion Multiplier (the PrusaSlicer field). 1.0 is no-correction.	Area-Cube: $\{1, 1.5, 2\}$; S-Test: $\{1, 1.57, 2, 2.5\}$. The value 1.57 matches the Cross-model prediction $1/k_{\text{flow}}$ for SF5.5 (NE-loaded CMC).

Table 2: Filename tokens. Example: Vp10-Fr1,5.gcode $\Rightarrow v_{\text{print}} = 10$ mm/s, EM = 1.5.

SAFETY / DATA INTEGRITY

“Vp” here is **print-head speed, not piston velocity**. In the rheology / simulation pipeline (CLAUDE.md, MATLAB run_solver_v3.m), V_p denotes *piston advance velocity* (typically ~ 0.01 mm/s for DIW). The G-code filenames in Impressao 3D/ use Vp as shorthand for v_{print} , the slicer’s perimeter speed (5–30 mm/s). Both quantities are physically distinct and orders of magnitude apart. Always read the gcode’s F(value) feedrate to confirm: F600 =

10 mm/s = Vp10.

4 PrusaSlicer configuration

The fixed-field profile from SOP v3.0 carries over unchanged (filament diameter 14.3 mm, retract 0, fan off, extrusion width 0.5 mm, etc.). Only the variable fields differ between G-codes within a sweep.

4.1 Fixed fields (one-time printer profile)

Slicer field	Value	Why
Filament diameter	14.3 mm	= $2R_{\text{syringe}}$, BD 10 mL. Makes 1 mm slicer- E = 1 mm piston. Critical.
Filament density	1.00 g/cm ³	Bookkeeping.
Cooling fan	OFF	DIW.
Retract length	0 mm	DIW.
Retract Z-hop	0 mm	DIW.
Travel speed	60 mm/s	Avoid head-drag through wet ink.
Skirt loops	1	Prime line.
Brim	0	DIW.
Top/bottom solid layers	0 (except Area-Cube top: 2)	Allow top-surface flatness reading on Area-Cube only.
Perimeters	1 (Hollow Cube, Scaffold, Tower) or 2 (Bridge)	Single perimeter for transparent failure-mode reading.
Extrusion width	0.5 mm	Matches 21G needle ID (0.515 mm).
First-layer extrusion width	0.5 mm	No squish.
First-layer speed	= Perimeter speed	Do not let PrusaSlicer slow the first layer.

Table 3: Fixed PrusaSlicer fields. Save in a printer + filament profile pair (e.g. DIW_BD10mL_21G).

4.2 Variable fields (per G-code)

Slicer field	Set to	Source / formula
Filament → Extrusion Multiplier	$EM = 1/k_{\text{flow,pred}}$	From <code>slicer_lookup*.csv</code> (output of <code>run_solver_v3.m</code>). For SF5.5: 1.57. For an unknown ink, start at 1.0 and refine.
Filament → Max volumetric speed	$Q_{\text{cap}} = \pi R_{\text{sy}}^2 V_p$	For BD 10 mL: Q_{cap} [mm ³ /s] = 160.6 · $V_{\text{p,piston}}$ [mm/s]. At $V_{\text{p,piston}} = 0.01$ mm/s, $Q_{\text{cap}} = 1.606$ mm ³ /s.
Print → Perimeter speed (and First-layer speed)	v_{print} from the test matrix	Sweep values: 5, 7, 10, 15, 30 mm/s (the “Vp” filename token).

Table 4: Per-G-code variable fields. All other parameters in Table 3 remain fixed.

4.3 Predicted EM (current ink panel)

4.4 Start / End G-code

PrusaSlicer-default homing is preserved in v4 (the manual M0 pause used in v3 is no longer required because each shape is run as a self-contained print). Confirm the slicer-generated header begins with M92 E2651.19 and G28; the head will return to the bed corner for the first move.

```
; Start G-code (PrusaSlicer default for DIW printer)
M92 E2651.19
G28 ; home all axes
```

Ink	$k_{\text{flow,pred}}$	$\text{EM} = 1/k_{\text{flow}}$	Notes
C15	0.83	1.20	Baseline CMC
C15 Gira 5.5 (SF5.5)	0.64	1.57	Nanoemulsion-loaded, lowest recovery
Bozano	1.06	0.94	Commercial reference

Table 5: Cross-model predictions, $V_{\text{p,piston}} = 0.01$ mm/s, 21G. The 1.57 value appears in the S-Test sweep as the SF5.5-tuned operating point.

```
G1 Z5 F5000 ; lift nozzle
G21 ; mm
G90 ; absolute coords
M82 ; absolute E
G92 E0
G1 E1 F3000 ; prime
G92 E0

; End G-code
G1 Z30 F600
M84
```

SAFETY / DATA INTEGRITY

If the printer’s mechanical Z-zero is set above the substrate (safety lift) rather than at substrate contact, jog Z down and issue **G92 Z0** *before* starting the print — the slicer assumes substrate contact at $Z = 0$.

5 Test matrix and execution order

5.1 Sweep grids

Shape	v_{print} (mm/s)	EM	Total G-codes
Area-Cube	5, 7, 10, 15, 30	1.0, 1.5, 2.0	15
S-Test	5, 7, 10, 15, 30	1.0, 1.57, 2.0, 2.5	20
Cubes (hollow)	5, 7, 10, 15, 30	1.0, 1.57, 2.0	15 (planned)
Tower-20mm	5, 10, 15	1.0, 1.57	6 (planned)
Tower-40mm	5, 10	1.0, 1.57	4 (planned)
Scaffold 0/90	5, 7, 10, 15, 30	1.0, 1.57, 2.0	15 (planned)
Bridge	5, 10, 15	1.0, 1.57	6 (planned)
Misc	per-shape (single best-EM scout)	1.57 (or refined)	TBD

Table 6: Sweep grids per shape. “Planned” entries are under generation by the operator; the SOP is forward-compatible. The S-Test grid is the most exhaustive because the raster printability index is the headline metric in the literature.

5.2 Recommended execution order

- S-Test full grid (20 prints).** Establishes the $(v_{\text{print}}, \text{EM})$ window for the ink. The raster geometry is the most sensitive to misalignment of EM with the ink’s recovery, so it converges first.
- Area-Cube grid (15 prints).** Validates the S-Test-derived window for filled regions.
- Hollow Cube + Scaffold + Tower + Bridge** at the two or three operating points that survived S-Test scoring.
- Misc shapes** as final demonstration prints at the single best operating point.

6 Photography protocol

Every printed specimen is photographed *immediately after the print finishes* (before any relaxation) and *again at $t = 30$ min relaxation* (for shapes where slump matters: Tower, Scaffold, Bridge, Area-Cube). The intermediate window is not photographed.

6.1 Per print: two iPhone images

Image pair acquisition

1. Mount iPhone in a fixed clamp directly above the print bed, camera plane parallel to the substrate. Mark the mount position on the bench — it must be reproducible across all prints in a sweep.
2. Set the iPhone to **standard Photo mode** (not Portrait, not Live, not RAW unless Pro-RAW is consistently used). HDR may remain on *Auto* but must be consistent within a sweep.
3. Capture **image A: with flash**. Tap to focus on the specimen centre, do not adjust exposure compensation.
4. Capture **image B: without flash**, identical framing, same focus point.
5. Filename convention: `<shape>_Vp<N>_Fr<X>_<ink>_<bg>_<flash|noflash>.jpg`
 - Example: `S-Test_Vp10_Flow1,57_SF5.5_Black_flash.jpg`
6. Save the pair into the corresponding `Impressao 3D/<shape>/<ink>/<bg>/` subfolder.

KEY IDEA

Why a flash + no-flash pair. The flash image *kills shadows and saturates highlights*, exposing geometry boundaries with high contrast — ideal for edge detection. The no-flash image *preserves the natural reflection / opacity gradient*, exposing layer-to-layer surface defects — ideal for texture analysis. Together the pair carries more information than a single “optimised” shot, and an automated pipeline can choose channel-by-channel which image to use.

6.2 Framing and substrate

Aspect	Specification
Camera axis	Perpendicular to substrate (top-down) for Cubes, Area-Cube, S-Test, Scaffold; perpendicular to tower axis (side-on) for Tower-20/40 and Bridge.
Working distance	Fixed; record it on the run sheet. The image pipeline cannot recover absolute scale without either (a) a reference object in frame or (b) a recorded working distance + iPhone lens focal length.
In-frame scale reference	<i>Strongly recommended:</i> place a printed RefBar (10×1 mm, 0.4 mm tall) or a ruler segment alongside every specimen. This is the same RefBar used in v3.0; STL still in <code>OLD/CCC_v1.stl</code> .
Background	Matte black or matte blue card, large enough to fill the field of view. Avoid glossy substrates — the flash returns a specular peak that corrupts edge detection.
Ambient lighting	Consistent within a sweep. Overhead fluorescent OK, mixed daylight + tungsten not OK. Record on the run sheet.

Table 7: Framing rules. Working distance and lighting are recorded because the image pipeline (Section 7) operates in pixel space until a reference object provides mm-per-pixel calibration.

7 Image-analysis pipeline (in development)

The pipeline ingests the flash + no-flash pair and emits a single *printability value* per specimen, in pixel units until a scale reference is provided. The full algorithm is under development; this section records the design intent so that the photography protocol remains forward-compatible.

7.1 Inputs and outputs

- **Inputs.** Two JPEGs per print: `<...>_flash.jpg` and `<...>_noflash.jpg`, plus the gcode-derived metadata ($shape, v_{\text{print}}, EM, h_{\text{layer}}, ink, bg$).
- **Outputs (pixel domain).** Per shape, a vector of geometric defect metrics — e.g. for the S-Test: line-width mean and SD, raster spacing mean and SD, fraction of expected raster nodes resolved. For the hollow cube: perimeter deviation (px), corner radius (px). For towers: base-bulge width vs nominal (px). Pixel $\rightarrow$ mm conversion is deferred to a calibration revision the operator will supply.
- **Headline scalar.** Per shape, a single *printability value* $\in [0, 1]$ combining the shape-specific metrics. Aggregation rule TBD during the brainstorming phase.

7.2 Stages (intent, not implementation)

1. **Pair registration.** Align the flash and no-flash images pixel-wise (small mount drift between shots is expected).
2. **Background segmentation.** Background colour (black or blue) is metadata; threshold-based segmentation produces a foreground mask.
3. **Reference-object detection.** If a RefBar or ruler is in frame, extract a px/mm scale; otherwise return pixel-only metrics.
4. **Per-shape metric extraction.** A shape-specific module (one per row of Table 1) computes the defect metrics.
5. **Score aggregation.** Per-shape printability value $\in [0, 1]$.

8 Measurement protocols (manual fallback)

Until the image-analysis pipeline is operational, each shape is scored manually. The protocols below are the human counterparts of the automated metrics in Section 7.

8.1 Hollow cube

Visually rate perimeter continuity (closed vs broken) and corner sharpness (sharp / rounded / collapsed) on a 0–3 scale. Photograph top-down; measure inner and outer edge length with caliper; deviation from nominal is the headline number.

8.2 S-Test / raster

Photograph top-down. Count the number of expected raster turn-around nodes that are resolved (i.e. where the strand clearly reverses without fusing into the previous pass). Printability index $= N_{\text{resolved}} / N_{\text{expected}}$.

8.3 Area cubes (different heights)

For each height, measure the top-surface gap fraction (fraction of the patch where adjacent raster lines have not fused) on a top-down photograph. Caliper the total stack height vs $N_{\text{sliced}} \cdot h_{\text{layer}}$ for the stacking ratio (carries over from v3.0 C3).

8.4 Tower 20 mm and 40 mm

Side-on photograph during and after the print. Record the layer count at which base width first exceeds nominal by 10% (N_{exp}). Compare to the SAOS prediction in `printing_parameters_per_ink.csv`.

8.5 Scaffold 0/90

Top-down photograph after relaxation. Count the fraction of pores that remain open (no strand sag bridging the pore). Pore closure rate is the headline metric.

8.6 Bridge

Side-on photograph. Measure mid-span sag distance with caliper against a known span length. Sag fraction = sag / span.

8.7 Misc shapes

Single qualitative verdict per shape: *pass* (geometry recognisable, no major defects) / *partial* (recognisable but with one major defect) / *fail*.

9 Run sheet

Per-Shape Print Run Sheet	Date: _____	Operator: _____
Print metadata		
Shape: _____ G-code file: _____		
v_{print} = _____ mm/s EM = _____ h_{layer} = _____ mm		
Ink: _____ Batch: _____ Needle: ☐ 21G ☐ 22G		
Substrate background: ☐ Black ☐ Blue ☐ _____		
Pre-print		
Filament diameter set to 14.3 mm? ☐ Yes ☐ No (STOP)		
Z-zero set at substrate contact? ☐ Yes		
Photography (at end of print)		
Mount position reproduced? ☐ Yes		
Working distance: _____ cm Ambient lighting: _____		
Image A (flash) saved? ☐ Image B (no flash) saved? ☐		
t = 30 min repeat pair (if applicable)? ☐		
Manual verdict (until imaging pipeline live)		

10 Deprecation notice

The Consolidated Calibration Coupon workflow described in SOP v3.0 (Latex/Printing_Parameters_SOP_v3_EM and CCC_v1.stl) is **deprecated as of May 19, 2026**. v3.0 and the legacy G-codes under Impressao 3D/OLD/ remain in the repository for archival reference. The active workflow for all new prints is the per-shape suite defined in this document.

11 Changelog

v4.0 (May 19, 2026). Replaced CCC_v1 single-coupon workflow with per-shape benchmark suite (hollow cube, area cubes, S-test, tower 20/40, scaffold 0/90, bridge, misc). New iPhone flash + no-flash photography protocol with shape/ink/background subfolder hierarchy. Sweep grids over (v_{print} , EM) documented per shape. Image-analysis pipeline placeholder added pending algorithm finalisation.

v3.0. CCC_v1 consolidated coupon with C1–C4 metrics on one specimen.

v2.0. 18 hand-coded G-codes per ink (C1–C4 battery).